"Annihilation of Matter"

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Introduction

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1 Electron-Positron Annihilation in the CRM Paradigm: Kinetic Energy of Core Collision Instead of Mass Deficit

The standard interpretation of electron-positron (e^-e^+) pair annihilation is based on Einstein's equation $E = m_e c^2$, where $m_e = 9.109 \times 10^{-31}$ kg is the rest mass of the electron, and c is the speed of light. The detected gamma photon with an energy of 511 keV is treated as a direct consequence of the **disappearance of the mass** of both particles and its conversion into radiation energy.

In this chapter, we will demonstrate that within the framework of the CRM paradigm (*Charge Regulation Model* – MRG), e^-e^+ annihilation can be explained **without resorting to the concept of mass deficit**. The 511 keV energy arises as a result of the **head-on collision of the massive cores** of the electron and positron, which were previously accelerated by their mutually attracting electric fields. The role of the constant c is reduced to imposing an upper limit on the speed of field reconfiguration, rather than being a result of the relativistic nature of spacetime.

1.1 CRM Axioms for the e^-e^+ System

We transfer the three axioms formulated for nuclear physics to a system of two particles with opposite charges:

1. **Mass as Field Inertia:** The electron mass m_e is a measure of the inertia of the configured electric field, rather than a primary source of interactions. The binding energy (here: annihilation energy) is the energy associated with the configuration of the charge fields.

2. **Non-linear Response of the Charge Field:** In systems with high charge density (such as the e^-e^+ pair at the moment their fields come into contact), the field exhibits a non-linear response. This leads to a logarithmic modification of the effective potential energy:

$$E_{\text{pot}}^{\text{CRM}}(r) = \frac{e^2}{4\pi\varepsilon_0 r}\left[1 - \gamma_c \ln\left(1 + \frac{r_e}{r}\right)\frac{1}{c^2}\right],$$

   where

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.8179 \times 10^{-15}\ m$$

   is the classical electron radius, and γ_c is the universal CRM regulator constant ($\gamma_c \approx 32.5$), determined by fitting to atomic nuclei masses.

3. **The Constant c as a Field Dynamics Regulator:** In systems

with high charge density, the speed of light constitutes the upper limit for the speed of field reconfiguration. This introduces an inertial cost proportional to $1/c^2$ in the field equations. Consequently, the kinetic energy of the cores cannot exceed $m_e c^2$.

2 Energetic Verification of the CRM-c Model for the $e^- e^+$ Pair

To prove that the energy released during annihilation originates from the work of the electrostatic field (converted into kinetics) rather than from the dematerialization of mass, we will perform an energetic analysis at the critical proximity distance $r = r_e$.

2.1 Step 1: Defining the Proximity Point

We assume that the moment of field "collision" occurs when the particles reach a distance equal to the classical electron radius r_e, defined as:

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \approx 2.8179 \times 10^{-15}\,\mathrm{m}$$

From this definition, a force dependency directly follows, which will serve as the foundation for further calculations:

$$\frac{e^2}{4\pi\varepsilon_0 r_e} = m_e c^2 \approx 0.511\,\mathrm{MeV}$$

2.2 Step 2: Calculation of the CRM-c Potential Energy

We substitute the distance $r = r_e$ into the non-linear CRM-c potential equation:

$$E_{\mathrm{pot}}^{\mathrm{CRM}}(r_e) = \frac{e^2}{4\pi\varepsilon_0 r_e}\left[1 - \gamma_c \ln\left(1 + \frac{r_e}{r_e}\right)\frac{1}{c^2}\right]$$

Simplifying the logarithmic term:

$$\ln\left(1+\frac{r_e}{r_e}\right)=\ln(2)\approx 0.6931$$

2.3 Step 3: Determining the Energy Value at Collision

By substituting the value $\gamma_c \approx 32.5$ (the constant derived from nuclear masses) and knowing that the term before the brackets for distance r_e is exactly $m_e c^2$, we obtain:

$$E_{\text{pot}}^{\text{CRM}}(r_e)=m_e c^2\left[1-\frac{32.5\cdot 0.6931}{c^2}\right]$$

2.4 Conclusions from the Calculations

1. **Field Energy as the Source:** The value of the Coulomb potential energy at distance r_e is identical to the rest energy of the electron (0.511 MeV). This means the field possesses a sufficient energy reserve to impart momentum corresponding to the particle's rest mass just before the collision. 2. **The Role of the Regulator γ_c:** The non-linear term $\gamma_c \ln(2)$ acts as a field saturation, preventing energy from escaping to infinity. 3. **Interpretation of Annihilation:** The total energy of the e^-e^+ pair (approx. 1.022 MeV) released during the encounter is, in this model, the **sum of the work of electrostatic fields** that has been converted into the kinetic energy of the collision.

Thus, gamma photons are not the result of the "disappearance" of mass m_e, but rather a sudden emission of kinetic energy that the particles gained by "falling" toward each other in a deep electrostatic potential well. The mass (the core) survives the collision but loses its field signature (charge), becoming an invisible neutral background to the instrumentation.

2.5 The Annihilation Mechanism in CRM

Consider an e^-e^+ pair in a vacuum. Each particle has:

- a massive core with mass m_e and radius $r_0 \ll r_e$,

- a spherical electric field extending to a distance $R = r_e/2$ from the center (which corresponds to the field range previously proposed based on annihilation energy calculations).

The annihilation process proceeds as follows:

1. When the distance between the centers of the cores reaches $2R = r_e$, the electric fields of both particles touch.

2. Electrostatic attraction accelerates the cores, and within a very short time (the so-called field relaxation time, of the order of R/c), the Coulomb force drops to zero.

3. At the moment of total field disappearance, the cores are at a distance $r_{\text{contact}} = r_e$ (centers) and already possess a certain "initial" velocity.

4. From this point on, the cores move solely under the influence of mutual gravity (which is enhanced at the subnuclear scale by the logarithmic density regulator, but in the simplest approximation, it can be neglected since the main contribution to kinetic energy comes from the work of electrostatic forces before their disappearance).

5. The cores collide head-on. The kinetic energy of the system in the center of mass just before the collision is equal to the work of the electrostatic forces over the path from $r = \infty$ to $r = r_{\text{contact}}$, i.e., the standard Coulomb potential energy:

$$E_{\text{kin}} = \frac{e^2}{4\pi\varepsilon_0 r_{\text{contact}}}.$$

   Substituting $r_{\text{contact}} = r_e$, we obtain:

$$E_{\text{kin}} = m_e c^2 = 511 \text{ keV}.$$

6. The kinetic energy of the collision is converted into a gamma photon of the same energy. γ_c enters the calculation as a negligible correction (on the order of 10^{-16}), because γ_c/c^2 is extremely small. Therefore,

current PET experiments are unable to distinguish the CRM model from the standard interpretation.

2.6 Comparison with the Standard Model

In the Standard Model:

$$E_\gamma = m_e c^2 \quad \text{(from mass deficit).} \tag{1}$$

In the CRM model:

$$E_\gamma = \frac{e^2}{4\pi\varepsilon_0 r_e}\left[1 - \gamma_c \ln(2)\frac{1}{c^2}\right] \approx m_e c^2 \left(1 - 2.5 \times 10^{-16}\right). \tag{2}$$

Both values are identical within the limit of current measurement precision ($\sim 10^{-5}$ eV). The difference is of **conceptual** importance, rather than empirical.

2.7 Falsifiable Prediction

Although both models yield the same energy for annihilation in a vacuum, the CRM paradigm predicts that this energy may **depend on the density of the medium** in which the annihilation occurs. In the atomic nucleus, the argument of the logarithm in the Coulomb regulator was Z/A. For an e^-e^+ system in matter, the natural argument is the local charge density ρ_q, normalized to a certain reference value $\rho_{q,0}$:

$$E_\gamma^{\mathrm{CRM}}(\rho_q) = m_e c^2 \left[1 - \gamma_c \ln\left(1 + \frac{\rho_q}{\rho_{q,0}}\right)\frac{1}{c^2}\right].$$

For low-density matter (e.g., gas), $\rho_q \ll \rho_{q,0}$ and the energy is close to $m_e c^2$. For high-density matter (metals, solids), a systematic, albeit small, shift in annihilation energy is expected, which could be tested in precision gamma spectroscopy experiments.

2.8 Summary

In the CRM paradigm, e^-e^+ annihilation is not evidence of $E = mc^2$, but rather an illustration of the fact that:

- mass is the inertia of the charge field,
- annihilation energy originates from the kinetic collision of massive cores,
- the constant c acts as a universal regulator of field dynamics.

The predicted difference between CRM and the standard model for vacuum annihilation lies below currently achievable precision, whereas the predicted dependence on medium density opens new possibilities for experimental testing of the theory.

3 Annihilation as a Process of Mechanical Defragmentation

The modern interpretation of electron-positron and proton-antiproton pair annihilation is based on the assumption of a total conversion of rest mass into electromagnetic radiation energy, according to the equation $E = mc^2$. However, analysis within the CRM-c paradigm indicates a fundamental energetic inconsistency in this model.

3.1 The Kinetic Energy Problem

In the process of approaching particles with opposite charges, Coulomb forces perform work, which imparts immense kinetic energy to the particles just before the moment of collision.

- The standard model assumes that the observed photon energy (511 keV for an electron) originates from the "disappearance" of mass.
- The question arises: what happened to the kinetic energy gained during the particles' fall toward each other?

According to the principle of conservation of energy, the sum of the energy from the alleged mass disappearance and the kinetic energy should be twice as high as what is observed. The absence of this surplus suggests that the observed photon energy is solely the result of the emission of kinetic energy (the "clatter" of the collision), while the mass of the cores is preserved.

4 Final Conclusions

Annihilation is not an act of destruction of matter, but an act of **reduction of electrostatic fields**. At the moment opposite charges meet, their spheres of interaction cancel each other out (destructive interference), releasing the cores from the corset of field interactions. The kinetic energy of the collision breaks the complex cores into smaller fragments (mass debris), which, due to the lack of a field signature (charge), become difficult to detect.

Matter does not disappear. Matter merely "loses its voice" in the electromagnetic spectrum.

Proton-antiproton annihilation provides definitive proof of the survival of matter. The observed secondary products, such as:

1. Pions ($\pi^{\pm}, \pi^0$),
2. Kaons (K),
3. Hadrons and leptons (neutrons, muons, neutrinos),

possess measurable rest mass. If mass still exists in the form of secondary particles after the "annihilation" process, then the claim of its total conversion into massless energy is false.

5 Scaling the CRM-c Model: Proton-Antiproton Annihilation

The transition from the electronic to the protonic scale requires accounting for an identical charge e with a rest mass approximately 1836 times greater.

In the CRM-c paradigm, this means that the work of the electrostatic field must be performed over a much shorter distance to generate kinetic energy corresponding to the rest mass of the proton.

5.1 Step 1: Determining the Critical Proximity Radius for the Proton

According to the model's logic, the rest energy of the proton $m_p c^2 \approx 938.27\,\mathrm{MeV}$ must be balanced by the field energy at the point of critical proximity r_p. We define the electrostatic radius of the proton analogously to the classical electron radius:

$$r_p = \frac{e^2}{4\pi\varepsilon_0 m_p c^2} \approx 1.5347 \times 10^{-18}\,\mathrm{m}$$

It is worth noting that r_p is three orders of magnitude smaller than r_e, which corresponds to the extreme concentration of inertia within the proton's core.

5.2 Step 2: Potential Energy Balance of the $p\bar{p}$ Pair

Substituting $r = r_p$ into the CRM-c equation, we obtain:

$$E_{\text{pot}}^{\text{CRM}}(r_p) = \frac{e^2}{4\pi\varepsilon_0 r_p}\left[1 - \gamma_c \ln\left(1 + \frac{r_e}{r_p}\right)\frac{1}{c^2}\right]$$

Note the significant difference in the logarithmic term. Since $r_e/r_p \approx 1836$, the logarithm takes the value:

$$\ln(1 + 1836) \approx \ln(1837) \approx 7.51$$

5.3 Step 3: Nonlinearity Analysis at the Protonic Scale

By substituting the regulator constant $\gamma_c \approx 32.5$, we obtain the nonlinear modification:

$$E_{\text{pot}}^{\text{CRM}}(r_p) = m_p c^2\left[1 - \frac{32.5 \cdot 7.51}{c^2}\right]$$

5.4 Conclusions for the Hadronic Scale

1. **Identity of the Mechanism:** The identity of the proton's and electron's charge means that the attraction mechanism is the same; however, due to the enormous mass density of the proton, Coulomb forces "work" down to distances of the order of 10^{-18} m. 2. **Extreme Kinetics:** The work of the field performed over this distance imparts kinetic energy of the order of 938 MeV to each particle core. The total collision energy of the $p\bar{p}$ pair is thus approximately 1876 MeV (3.3×10^{-10} J). 3. **Defragmentation instead of Annihilation:** At such gigantic kinetic energy, the collision cannot be "clean." The proton core (composed of hundreds of neutral packets) undergoes mechanical shattering into pions and kaons.

These calculations prove that the 1876 MeV energy observed during proton annihilation is solely the result of the work of the electrostatic field converted into momentum, rather than the dematerialization of mass. The energy scale increases because the proton core allows for a much deeper "fall" into the potential well than in the case of the electron.

6 Analysis of $p\bar{p}$ Collision Products as Evidence of Mass Defragmentation

Empirical analysis of the processes occurring during proton-antiproton collisions demonstrates that so-called "annihilation" does not lead to the total disappearance of matter. On the contrary, detectors record a wide spectrum of products with specific rest masses, suggesting that the collision energy (originating from the work of the field) serves to mechanically shatter complex cores into smaller fragments.

6.1 Catalog of Collision Products (the so-called Mass Debris)

Listed below are the primary products recorded as a result of collisions, which de facto constitute shards of the shattered nuclear structure:

1. **Gamma Radiation:** High-energy photons resulting from the sudden emission of kinetic energy at the moment of field reduction (the equivalent of a shock wave).

2. **Pions (π^+, π^-, π^0):** The most common nuclear fragments. Neutral pions rapidly decay into photons, which is often misinterpreted as evidence of the primary masslessness of the process.

3. **Kaons (K):** Heavier hadronic particles produced at higher collision energies.

4. **Complex Hadrons:** Baryons (e.g., neutrons, hyperons) and mesons, which are larger clusters of neutral cores that survived the shattering process.

5. **Leptons (muons μ, neutrinos ν):** Decay products of unstable fragments, constituting the finest, hard-to-detect mass dust.

6. **Quarks and Gluons:** Occurring at extreme collision energies; they serve as evidence that the degree of fragmentation of matter is a function of the delivered kinetic energy.

6.2 Dependence of Fragmentation on Collision Energy

A key argument for the mechanical nature of this process is the fact that **annihilation results depend directly on the collision energy**. In the classical $E = mc^2$ model, the outcome should be constant (energy from mass). However, in reality, the higher the kinetic energy of the collision, the deeper the defragmentation:

- At low energies, we primarily obtain pions.
- At extreme energies, the core shatters into quarks and gluons.

This proves unequivocally that mass is not annihilated but undergoes **mechanical fragmentation**. These particles are visible in detectors only as long as they possess high momentum. Upon losing kinetic energy (decelerating), they lose the ability to interact with measuring equipment and become part of the invisible, neutral background.

7 Demystification of Antimatter and Energy Balance Errors

Modern particle physics assigns antimatter the status of an exotic form of existence that undergoes total annihilation upon contact with matter. However, an analysis of production and annihilation processes within the CRM-c paradigm suggests a completely different interpretation: antimatter is not a distinct type of substance, but merely **matter with a forced, unnatural polarization of the electrostatic field.**

7.1 Artificial Polarization of the Core

The process of antiproton production in accelerators (e.g., during proton-target collisions) can be interpreted as a rare statistical event of a mechanical nature.

- A proton is not "created from energy"; rather, during an extremely energetic collision, its internal structure (positive charge) is disrupted.
- In place of the removed positive field phase, a negative (electronic) phase is "pressed in" as a result of the collision's chaos.
- The resulting object possesses the same massive core, but its interaction with the environment is reversed. This is a polarization error, not a new substance.

7.2 Energy Accounting as an Interpretational Camouflage

The most controversial aspect is the method of calculating the energy balance of annihilation. Official reports claim consistency with the $E = mc^2$

equation; however, this is achieved through a specific mathematical maneuver:

$$E_{obs} + \sum(m_{deb}c^2) = E_{tot} \tag{3}$$

Where:

- E_{obs} – observed gamma radiation energy (the "clatter" of the collision),
- $\sum(m_{deb}c^2)$ – energy back-calculated from the mass of the recorded "mass debris" (pions, kaons, etc.).

Critique of the Method: Physicists do not measure the pure energy of mass annihilation; instead, they sum the energy released with the mass that survived the collision, converting that mass back into energy. This is a circular argument. If the mass were truly annihilated, no massive fragments should appear in the balance ($m_{deb} = 0$). The presence of "mass debris" with high inertia is direct evidence that the original matter did not disappear, but merely underwent fragmentation and lost its field signature of charge.

7.3 Conclusions

The claim of "pure annihilation energy" is a myth resulting from a misinterpretation of measurements. Antimatter is matter with a "wrongly fitted" charge, and the process of its encounter with matter is a field short-circuit, leading to a powerful kinetic collision of the cores. Instead of investigating the mechanics of this decay, physics hides it behind the screen of Einstein's equation, adding the mass of the fragments to the energy balance to maintain the dogma of disappearing matter.

7.4 The Detection Problem and "Missing" Mass

The key reason why modern physics erroneously assumes the total disappearance of mass in the annihilation process is the limitation of detection methods. Most research apparatus (including mass spectrometers) rely on the interaction of external fields with the **electric charge** of the studied particle.

Neutral particles, which are products of core defragmentation, behave analogously to the neutron:

- Lacking a net charge, they do not undergo deflection in magnetic fields.
- They do not initiate classical electromagnetic cascades in detectors.
- Their presence can only be demonstrated indirectly, through mechanical collisions with other nuclei, which is often interpreted as "noise" or measurement error.

The fact that after a proton-antiproton collision we record **mass in the form of pions and other hadrons** is definitive proof that mass is not "fuel" for photons, but a permanent building block that has simply lost its field-based "visibility" (charge, momentum). If annihilation truly destroyed matter, a $p\bar{p}$ collision could not yield any products with rest mass – yet it produces a whole spectrum of them.

7.5 Ontological Summary

In light of the above, Einstein's equation $E = mc^2$ should be interpreted not as a recipe for "creating energy from matter," but as a **description of the maximum kinetic energy** that an electrostatic field can impart to a mass of a given density before its mechanical shattering (an upper limit resulting from c).

Annihilation is not the death of matter – it is its transition to a neutral state. A state that eludes detectors built exclusively to "hear" electric charge.

Within the FBP (Photon Without Momentum) and CRM (Charge Regulation Model) paradigms, this process is consistent with the conservation of mass, energy, and charge – without the need to resort to the hypothesis of total conversion of matter into pure energy. It is not matter that disappears – it merely "loses its voice" in the electromagnetic spectrum.

Mieczyk

Author of the "Annihilation of Matter" Theory.